

SAI KRISHNA MATETI

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PROFESSIONAL SUMMARY

Backend Engineer and Node.js Developer with 4.9+ years of experience building and scaling RESTful APIs, microservices, and event-driven architectures for a cloud-native healthcare platform (Node.js, Express.js, MongoDB, Redis, Elasticsearch, Azure). Delivered 120+ API endpoints across 6 microservices serving 50,000+ users — patients, doctors, pharmacies, and labs. Deep expertise in API security (JWT, OAuth2, RBAC), rate limiting, load balancing, caching strategies, and CI/CD pipelines (Docker, GitHub Actions). Reduced API 95 response time by ~50% through Redis caching and MongoDB index tuning. Proven ability to write clean, testable code (Jest, Supertest) and deliver production-grade Node.js backends with structured logging, error monitoring, and zero security incidents over 2+ years.

TECHNICAL SKILLS

Languages:	JavaScript (ES6+), TypeScript (interfaces, generics, decorators)
Frontend:	React.js (Hooks, Context API, Redux), HTML5, CSS3, REST API integration
Backend & APIs:	Node.js, Express.js, REST APIs, WebSockets, Microservices, Event-Driven Architecture, Async/Await, API Versioning, API Gateway, Rate Limiting, Request Throttling, Cron Jobs
Testing:	Jest, Supertest, unit testing, integration testing, API testing
Auth & Security:	JWT, OAuth2, RBAC, Multi-tenant access control, Secure signed URLs
Databases & Caching:	MongoDB (aggregation pipelines, indexing), MySQL, Redis, Elasticsearch
Cloud & DevOps:	Microsoft Azure (Blob Storage, NLP/Text Analytics, Service Bus, Message Queue), Docker, GitHub Actions (CI/CD), Nginx (reverse proxy, load balancing), Scalable Architecture, High Availability
Tools & Observability:	Postman, Swagger/OpenAPI, Git, Linux, Structured Logging, Error Monitoring, Alerting, Performance Tuning, Code Review, Agile, Scrum

PROFESSIONAL EXPERIENCE

Software Engineer L3 | Archents IT Pvt Ltd

Nov 2021 – Present

- Architected and delivered 120+ RESTful API endpoints across 6 microservices (patient, doctor, pharmacy, lab, payments, notifications) using Node.js and Express.js; implemented API rate limiting, request throttling, and input validation to protect services under high load.
- Improved average API response time by ~50% by introducing Redis caching layers, query optimisation, and compound MongoDB indexes on high-frequency collection scans; applied performance tuning and profiling to identify bottlenecks.
- Configured Nginx as a reverse proxy and load balancer across microservice instances, ensuring even traffic distribution, high availability, and fault tolerance; contributed to 99.9%+ uptime over 2+ years.
- Built scalable, event-driven workflows using Azure Service Bus (message queue) processing 10,000+ async events/day with 99.9% delivery reliability; used async/await and non-blocking I/O patterns throughout; integrated Azure NLP/Text Analytics cutting manual data-entry by ~30%.
- Implemented JWT/OAuth2 + RBAC with multi-tenant data isolation across 5 user roles; integrated Azure Blob Storage handling 1,000+ file uploads/downloads per day with secure signed-URL access.
- Containerised microservices with Docker and built GitHub Actions CI/CD pipelines, cutting release cycle from 2 days to under 4 hours; standardised deployments eliminated 3 production outages in 6 months.
- Developed React.js admin dashboards for internal operations — appointment management, pharmacy inventory, and lab report tracking — integrated directly with backend microservices via REST APIs.
- Wrote Jest and Supertest unit/integration tests achieving >80% coverage on critical service modules; enforced testdriven quality gates in CI pipeline.
- Wrote Swagger/OpenAPI specs for all 40+ endpoints; led code reviews, pair programming sessions, and sprint planning in an Agile/Scrum SDLC with weekly releases.
- Implemented automated backup schedules, structured logging, alerting pipelines, and error monitoring, reducing mean time to incident detection by ~40%.

Key achievements:

- 50% reduction in API 95 response time via Redis + index tuning — doctor dashboard load improved from ~1.8s to <900ms.
- Standardised Docker deployments eliminated 3 environment-related production outages in the 6 months following rollout.
- Automated backup and monitoring pipelines cut incident detection time from 45 min average to under 10 min.

PROJECTS

Ciana Healthcare Platform (Microservices Architecture) — Backend Developer

- Patient services: built registration, profile, appointment booking, wellness tracking, Face Scan, Vitals Details, and prescription upload flows.
- Payment integration: implemented Razorpay and PhonePe payment gateways for 3 package types (consultations, lab tests, wellness plans), handling ₹2M+ in monthly transactions.
- Doctor workflows: developed video-consultation scheduling, EMR generation, digital prescriptions, and end-to-end appointment lifecycle; supported 500+ doctor–patient sessions per day.
- Face Scan AI module: async result-processing pipeline via Azure Service Bus, cutting vitals-capture time from 5 min to under 60 seconds.
- Pharmacy & Lab: inventory management with fuzzy medicine search, auto purchase-order generation on low stock; lab scheduling, PDF report upload to Azure Blob with WebSocket real-time notifications.
- Security & compliance: enforced RBAC, JWT auth, and data-access boundaries across all services; zero security incidents in 2+ years of production.

ACADEMIC PROJECTS

Crime Management System — Major Project (B.Tech)

- Designed and developed a full-stack web application to digitise FIR registration, case tracking, and criminal record management for police departments, replacing manual paper-based workflows.
- Built role-based modules for Admin, Officer, and Citizen with secure login and access control, enabling differentiated data visibility and case-update permissions.
- Implemented CRUD operations for case files, suspect records, and evidence logs using a relational database, with search and filter functionality across case status, location, and crime type.
- Designed normalized database schema to manage relationships between cases, officers, and suspects, ensuring data integrity and efficient querying.

Student Attendance Management System — Mini Project (B.Tech)

- Built a web-based system to automate daily student attendance tracking for a college, replacing manual register-based attendance with a centralised database-driven workflow.
- Developed modules for faculty to mark attendance, generate subject-wise and student-wise attendance reports, and flag students below the minimum attendance threshold.
- Designed database schema to store student, course, and attendance records, with queries to calculate attendance percentage per subject and per semester.

EDUCATION

B.Tech – Computer Science

SVS Group of Institutes, Hanamkonda, Telangana

2017 – 2020

Diploma – Computer Engineering

Government Polytechnic College, Warangal, Telangana

2014 – 2017